

STAT 301 In-class Activity No. 5

Consider the “wage” dataset again. Now we fit a multiple linear regression model, with WAGE as the response and **all** other variables in the dataset as predictors. No need to consider interactions.

- (a) Is there a potential multi-collinearity problem among the continuous predictors (check pairwise correlations, VIF is optional)?
- (b) Any predictors may be removed from the model?
- (c) Is this an observational study or designed experiment? Can we claim causal relationship between the significant predictors and the response?